

Reflective Lantern - Innovation Report

New Project Built

Name: Cart-Mind

URL: <https://github.com/atharvadevne123/reflective-lantern/tree/innovation/cart-mind>

Date:

Inspired by: Portfolio gap analysis: recommender systems and purchase-intent ranking were the largest untouched ML domain (HN and GitHub trending unreachable - egress policy blocks non-proxied HTTP).

What it does: Real-time product recommendation and purchase intent prediction API using a LightGBM/XGBoost/RandomForest soft-voting ensemble, FAISS item-similarity search, KS-test and PSI drift monitoring, and champion/challenger retraining.

Mandatory Tech Stack Coverage: 100%

- [X] Python 3.11
- [X] FastAPI (9 versioned endpoints)
- [X] Ensemble: LightGBM + XGBoost + RandomForest
- [X] sklearn Pipeline (5 stages, 29 features)
- [X] Model monitoring (KS + PSI drift, DB logging)
- [X] Docker + docker-compose + PostgreSQL
- [X] SQLAlchemy (SQLite dev, PostgreSQL prod)
- [X] pytest suite (186 tests, 9 modules)
- [X] .env.example
- [X] GitHub Actions CI (ruff + pytest + wheel)

Optional Enhancements (7 of 4 required)

- [X] RAG + FAISS similarity index
- [X] Airflow retraining DAG
- [X] Automated retraining (champion/challenger)
- [X] Drift detection (KS-test + PSI)
- [X] 5-fold CV + AUC-ROC
- [X] Alembic migrations
- [X] TTL cache + rate limiting + correlation IDs

Project Stats

Files created: 36

Tests written: 186 (all passing)

Commits: 61

Model CV AUC-ROC: 0.717 +/- 0.019 (5-fold stratified, 5000 samples)

Ruff lint + format: clean

Defects Found and Fixed During Build

Champion compared against itself

`train_model` writes `metrics.json` as a side effect, and `retrain_task` read the champion AUC from that file after training - loading the challenger's own metrics. The gate was `'auc >= auc'`, always true, so every challenger was promoted including regressions. Champion is now read before training; a rejected run restores the champion's metrics.

Ad-hoc index builds clobbered the serving FAISS index

`build_faiss_index` wrote to the shared index path unconditionally, so any caller building a throwaway index replaced the deployed one - leaving a mismatched-dimension index that failed inside FAISS on an opaque assertion. Persistence is now opt-in; `load_faiss_index` rejects a dimension mismatch and rebuilds; `search_similar_items` raises a clear error.

Data Honesty Note

This repository ships no proprietary retail dataset. Training rows come from `make_sample_dataframe` and labels from `make_purchase_labels`, which samples outcomes from a logistic transform of a latent propensity score built from the behavioural drivers that genuinely move conversion (clicks, wishlist intent, category affinity, recency, discount depth, price friction). Gaussian noise keeps the problem non-separable, so the reported 0.717 AUC reflects a model learning a real but imperfect signal rather than a number inflated by a trivially separable target. An earlier bootstrap fit on random labels scored 0.524, as expected for pure noise.

Delivery Status

Standalone repo creation returned 403: this session's GitHub access is bound to the reflective-lantern repository. The project ships on branch `innovation/cart-mind` with a draft pull request. Outbound SMTP is unavailable (raw TCP to ports 465 and 587 times out; the egress proxy passes HTTPS only), so this report was committed to the repository rather than emailed.